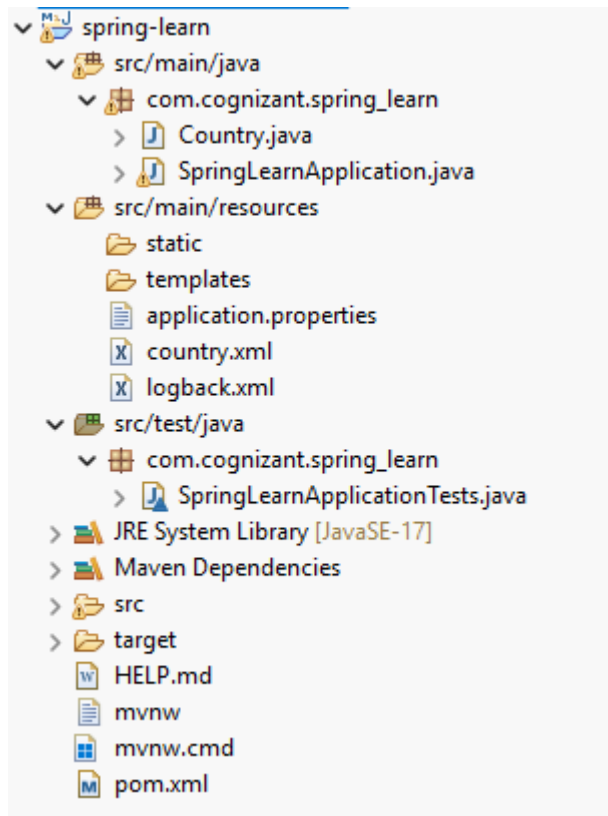


Spring Core – Load Country from Spring Configuration XML



pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.3</version>
    <relativePath/>
  </parent>

  <groupId>com.cognizant</groupId>
  <artifactId>spring-learn</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>spring-learn</name>

  <properties>
    <java.version>17</java.version>
  </properties>

  <dependencies>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
```

```

<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-devtools</artifactId>
  <scope>runtime</scope>
  <optional>true</optional>
</dependency>

<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>

<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-context</artifactId>
</dependency>

</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
</project>

```

application.properties:

```

spring.application.name=spring-learn
logging.level.root=INFO
logging.level.com.cognizant.spring-learn=DEBUG

```

country.xml:

?? xml	version="1.0" encoding="UTF-8"
▼ beans	(description?, (import alias bean namespace:uri="##other")*, beans*)
xmlns	http://www.springframework.org/schema/beans
xmlns:xsi	http://www.w3.org/2001/XMLSchema-instance
xsi:schemaLocation	http://www.springframework.org/schema/beans https://www.springframework.org/schema/beans/spring-beans.xsd
▼ bean	((description?, (meta constructor-arg property qualifier lookup-method replaced-method namespace:uri="##other")?))
id	country
class	com.cognizant.spring_learn.Country
▼ property	(description?, (meta bean ref idref value null array list set map props namespace:uri="##other")?)
name	code
value	IN
▼ property	(description?, (meta bean ref idref value null array list set map props namespace:uri="##other")?)
name	name
value	India

logback.xml:

▼ configuration	
▼ appender	
name	CONSOLE
class	ch.qos.logback.core.ConsoleAppender
▼ encoder	
pattern	%d{HH:mm:ss} [%thread] %-5level %logger{36} - %msg%n
▼ root	
level	DEBUG
▼ appender-ref	
ref	CONSOLE

Country.java:

```
package com.cognizant.spring_learn;

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

public class Country {

    private static final Logger LOGGER =

        LoggerFactory.getLogger(Country.class);

    private String code;

    private String name;

    public Country() {

        LOGGER.debug("Inside Country Constructor.");

    }

    public String getCode() {

        LOGGER.debug("Inside getCode()");

        return code;

    }

    public void setCode(String code) {

        LOGGER.debug("Inside setCode()");

        this.code = code;

    }

    public String getName() {

        LOGGER.debug("Inside getName()");

        return name;

    }

    public void setName(String name) {

        LOGGER.debug("Inside setName()");

        this.name = name;

    }

    @Override
```

```

    public String toString() {

        return "Country [code=" + code + ", name=" + name + "]";

    }

}

SpringLearnApplication.java:
package com.cognizant.spring_learn;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class SpringLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(SpringLearnApplication.class);

    public static void main(String[] args) {

        displayCountry();

    }

    public static void displayCountry() {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");

        Country country =
            context.getBean("country", Country.class);

        LOGGER.debug("Country : {}", country.toString());

    }

}

```

OUTPUT:

```

12:30:41 [main] DEBUG o.s.c.s.ClassPathXmlApplicationContext - Refreshing org.springframework.context.support.ClassPathXmlApplicationContext@11a9e7c8
12:30:41 [main] DEBUG o.s.b.f.xml.XmlBeanDefinitionReader - Loaded 1 bean definitions from class path resource [country.xml]
12:30:41 [main] DEBUG o.s.b.f.s.DefaultListableBeanFactory - Creating shared instance of singleton bean 'country'
12:30:41 [main] DEBUG com.cognizant.spring_learn.Country - Inside Country Constructor.
12:30:41 [main] DEBUG com.cognizant.spring_learn.Country - Inside setCode()
12:30:41 [main] DEBUG com.cognizant.spring_learn.Country - Inside setName()
12:30:41 [main] DEBUG c.c.s.SpringLearnApplication - Country : Country [code=IN, name=India]

```

Spring lifecycle

1. ClassPathXmlApplicationContext loads country.xml

↓

2. Spring sees <bean id="country" class="...Country">

↓

3. Calls new Country() → "Inside Country Constructor."

↓

4. Calls setCode("IN") → "Inside setCode()"

↓

5. Calls setName("India") → "Inside setName()"

↓

6. context.getBean("country") → returns the ready bean

↓

7. LOGGER.debug(country) → "Country [code=IN, name=India]"